

Joshua Dai

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EDUCATION

University of Massachusetts Amherst – Riccio College of Engineering

Expected May 2027

M.S. Electrical and Computer Engineering

B.S. Biomedical Engineering | Minor in Computer Science | GPA: 3.965

Graduated 2026

EXPERIENCE

Mechatronics Engineer

May 2026 - Present

Ginkgo Bioworks

Boston, MA

- Led E17 enclosure redesign enabling new instruments deployment across RAC systems
- Converted an outside firm's drawing set to Ginkgo standards, releasing 13 prints and bringing design in-house
- Assembled buffer nests, instruments, and RAC hardware, bringing 3 iKits online from install to Python driver
- Taught 6-DOF trajectories to under 0.05 mm, preventing collisions and dropped labware that halt runs
- Debugged mechanical, electrical, and controls faults on RAC platform protecting throughput
- Characterized inrush transients and current draw, defining rigorous FAT power criteria to resolve electrical faults

R&D Process Engineer

June 2025 – August 2025

Takeda Pharmaceuticals

Lexington, MA

- Engineered advanced Antibody-Drug conjugation protocols for manufacturing efficiency and product stability
- Leveraged TECAN robotics to execute 100+ automated test conditions, optimizing throughput by 60%
- Drove an 80% reduction in overall manufacturing costs and co-authored 2 peer-reviewed publications
- Achieved a 96% reduction in drug consumption by engineering a novel process that eliminated 3 unit operations

R&D Engineer

January 2024 – June 2024

Ocular Therapeutix

Bedford, MA

- Optimized Glaucoma ocular insert manufacturing parameters, achieving a 30% reduction in active drug waste
- Drove a 70% increase in material testing workflow by designing custom SolidWorks fixtures
- Accelerated data processing workflows by 5x by implementing Python automation using Pandas and NumPy

PROJECTS

Ouro Bot – Autonomous Trash-Detection Robot

September 2026 – Present

- Designing full mechanical assembly in Onshape for a tracked, autonomous household robot with a tilting head
- Integrating 2D LiDAR and Pi Camera for SLAM-based apartment mapping and real-time trash detection
- Developing a Raspberry Pi object-detection pipeline to classify household clutter and trigger alerts
- Compiling full BOM and sourcing plan across drivetrain, compute, and sensing subsystems ahead of assembly

Baystate Hospital – SafeStep

September 2025 – May 2026

- Architected ARM Cortex-M4 wearable fall prevention device integrating real-time 6-axis IMU and ECG data
- Deployed random forest classifier using MicroPython, triggering 800 PSI airbag deployment in <0.5s via solenoid
- Designed Buck Converter power stage to drive 3.7V high-current solenoids from 3.3V MCU logic
- Implemented on-device EMA low-pass filter and free-fall physics override (<3.0 m/s²) to reduce false negatives live

Mechanofluidic CNS Hemodynamic Modeling System

October 2025 – May 2026

- Pioneered 12-well cone-and-plate Onshape model to study low shear stress astrocyte differentiation
- Simulated micro-channel fluidics via COMSOL CFD to map complex flow regions inaccessible to physical sensors
- Validated computational CFD simulations against wet-lab assays to verify complex fluid-cell stress profiles

UMass Robotics – 6-DOF Rehabilitative Robotic Arm

August 2024 – April 2025

- Designed a cable-driven 6-DOF robotic arm for upper-limb rehabilitative assistance
- Engineered the mechanical assembly in Onshape, optimizing structural robustness and range of motion
- Conducted MSC Adams motion analysis targeting daily-task assistance to restore patient autonomy

PUBLICATIONS

Ni, K., Wang, L., Shupe, A., Cooney, T., **Dai, J.**, et al. "On-Column Thiol Protection Strategy to Control Charge and Stability of ThiomAb." *Antibody Therapeutics*, 2026.